

# Justin Chan

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## Summary

Software engineer who enjoys building software that connects practical product work with deep technical systems. UC Irvine graduate experienced in full-stack web applications, search infrastructure, and reinforcement learning — most recently training an autonomous game-playing agent from scratch. Looking for a role where I can contribute to a strong team, ship clean and maintainable systems, and keep growing as an engineer.

## Education

**University of California, Irvine**

**Sep 2021 - Mar 2025**

*Software Engineering*

- **GPA:** 3.459
- **Coursework:** Design and Analysis of Algorithms, Internet Applications Engineering, Data Structure Implementation and Analysis, Software Design: Structure and Implementation, Software Design: Applications, Human-Computer Interaction, Principles of Operating Systems, Information Retrieval, Project in Databases and Web Applications, Software Testing, Analysis, and Quality Assurance

## Projects

**AscensionAI** | <https://github.com/JustinoChan/AscensionAI>

**Nov 2025 - Present**

- Technologies: Python, PyTorch, Gymnasium, NumPy, Matplotlib
- Built a reinforcement learning system that trains an autonomous Ironclad agent against a live Slay the Spire process through ModTheSpire, CommunicationMod, and SpireComm.
- Implemented a Gymnasium-style environment wrapper with a 530-dimensional observation encoder, 134-action discrete space, legal-action masking, and shaped rewards.
- Developed a PyTorch actor-critic PPO training pipeline with behavior cloning warm-start, GAE returns, entropy annealing, and gradient clipping.
- Architected parallel rollout collection with atomic checkpoints, crash recovery, worker heartbeats, and stale-rollout rejection.

**Search Engine** | <https://github.com/Vincent10351/Indexer>

**Feb 2023 - Mar 2023**

- Technologies: Python, JSON, Algorithm Optimization
- Engineered a Python-based search engine that indexed and processed 56,000+ web pages, enabling rapid information retrieval across a large corpus.
- Developed a robust query handling system that improved search accuracy and interpretation of user intent.
- Reduced average query response time by 35% through custom tokenization and algorithm optimization strategies.

**Capstone Project Archive** | <https://github.com/cpark50/capstone-archive>

**Jan 2024 - Jun 2024**

- Technologies: React.js, JavaScript, Python, Django, MySQL8, Firebase
- Developed a Capstone Archive web application for hosting student projects.
- Implemented Firebase Authentication and MySQL8 Database integration.
- Contributed to frontend development with CSS and React.js.
- Modified Django REST API for data management.
- Established Agile workflows by implementing Jira boards, hosting weekly meetings, and coordinating daily standups via Discord to improve team communication and delivery.

**BitLink** | <https://github.com/SusLiu03/BitLink>

**Mar 2024 - Jun 2024**

- Technologies: Angular, TypeScript, MongoDB, Node.js, Passport, BCrypt
- Built and deployed a full-stack social media platform supporting real-time messaging, user authentication, and social interactions (posts, likes, follows).
- Developed scalable APIs in Node.js to handle data persistence for records in MongoDB.
- Implemented secure authentication using Passport.js and BCrypt, ensuring encrypted credential storage and session handling.
- Designed responsive UI in Angular + TypeScript, improving engagement and usability across devices.

## Skills

- **Programming Languages:** Java, Python, C, C++, JavaScript, TypeScript
- **Web Development:** HTML, CSS, AngularJS, React.js, Django
- **AI / Machine Learning:** PyTorch, Gymnasium, NumPy, Matplotlib, Reinforcement Learning
- **Data Management:** SQL, MongoDB, Firebase
- **Software Engineering:** Software Development, Software Testing, Algorithms, Design Principles, Design Patterns, API Integrations
- **Methodologies:** Agile/Scrum, DevOps Practices, Project Management